

ARAVIND VELAGA

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972-678-0442

Databricks Certified Data Engineer Associate

Dallas, TX

SUMMARY

Results-driven **Data Engineer with 8 years of experience** building scalable, high-performance data pipelines across **AWS, Databricks, and big-data ecosystems**. Expert in **Python, PySpark, ETL frameworks, data lake/lakehouse architectures**, and modern ingestion patterns including real-time streaming (Kafka/Kinesis) and batch processing. Proven ability to design reusable frameworks, optimize distributed systems, enforce strong data-quality practices, and deliver **analytics-ready & ML-ready datasets** for enterprise use cases in **healthcare, finance, and retail**.

Key strengths include:

- Production-grade Python ETL frameworks using design patterns (Factory, Lambda, Kappa)
- Optimizing Spark/Databricks jobs for partitioning, joins, cluster cost & runtime efficiency
- Building automated DQ checks, schema validation, reconciliation & audit frameworks
- Orchestrating workflows using Airflow, Step Functions, and CI/CD pipelines
- Strong experience across AWS Glue, EMR, Redshift, S3, CloudTrail & data governance

TECHNICAL SKILLS

Big Data & Processing: PySpark, Apache Spark, Databricks, Hadoop

Cloud Platforms: AWS (Glue, EMR, Redshift, S3, Kinesis, Lambda), Azure (ADF, Synapse)

Streaming & Orchestration: Kafka, NiFi, Airflow, AWS Step Functions

ETL & Databases: SQL, Snowflake, Oracle, SQL Server, Hive, Iceberg

Programming: Python, Shell Scripting, Terraform

Visualization: Power BI, Tableau, QuickSight, SageMaker

DevOps & Tools: Git, Azure DevOps, GitHub Actions, CI/CD, JIRA, Agile/Scrum

CERTIFICATIONS

- Databricks Certified Data Engineer Associate
- Databricks Fundamentals
- Data Analysis with Python
- Database & SQL for Data Science
- Python for Data Science & AI
- KNIME Certifications

PROFESSIONAL EXPERIENCE

Data Engineer – Centene Corp, St. Louis, MO

July 2024 – Present

- Developed scalable **ETL pipelines using AWS Glue, Talend, SSIS, and Python**, integrating Oracle, MongoDB, and SQL Server into **S3 + Redshift** for enterprise analytics.
- Built **Python-based DQ & validation frameworks** with schema checks, reconciliation logic, and automated compliance checks for Medicaid/MMIS datasets.
- Implemented **Databricks Auto Loader + Delta Lake** for clinical/EHR data ingest, reducing onboarding time and improving reliability.
- Designed **Python ETL frameworks** leveraging patterns like Factory & Kappa for real-time ingestion via **Kafka, Kinesis, and Glue**.
- Built **cross-account data sharing pipelines** using Lake Formation, enabling secure collaboration across Medicaid program teams.

- Designed **cost-optimized data lake zones** (raw → refined → curated), reducing S3 and compute cost.
- Tuned **Spark/EMR clusters**, partitioning strategies, and join optimizations to reduce pipeline runtime.
- Engineered large-scale **TPL & claims pipelines** delivering ML-ready data to SageMaker.
- Built CI/CD pipelines using **AWS CodePipeline & CodeBuild** for automated deployments.
- Implemented **CloudTrail + audit logging** for compliance and security monitoring.
- Designed KPI dashboards using **Power BI and QuickSight** to support cost analysis and trend insights.
- Created a **DQ rules engine** (Python + Spark) supporting 50+ configurable checks including null profiling, referential integrity, uniqueness, and statistical validation.
- Integrated **Great Expectations-style validation logic** into Delta pipelines, enforcing data SLAs across TPL & EHR datasets.
- **Implemented Glue Job bookmarks and incremental loading strategies** to optimize cost and avoid duplicate processing.
- Built **cross-account data sharing pipelines** using Lake Formation, enabling secure collaboration across Medicaid program teams.
- Re-architected heavy transformations using **broadcast joins, Z-ordering, optimized file sizes**, cutting runtimes.
- Converted high-cost Glue jobs into **serverless Databricks workflows**, improving scalability and reducing overhead.
- **Developed metadata-driven ingestion frameworks** enabling dynamic schema management and reusable components.
- **Configured Delta Live Tables (DLT)** to enforce data quality expectations and automated lineage tracking.
- Designed **Kinesis real-time ingestion framework** with checkpointing, idempotency, and DLQ handling for streaming events.
- Integrated **Kafka-based micro-batch pipelines** with Delta Lake for near real-time insights into claims and provider activity.
- **Optimized Redshift sort/distribution keys** improving query performance.
- **Automated EMR cluster lifecycle management**, reducing cost.
- Automated end-to-end ETL deployments using **Terraform + CodePipeline**, improving release reliability and reducing manual work.
- Implemented **automated rollback strategy** for failed ETL deployments to ensure zero downtime.
- **Built exception-handling and retry-logic frameworks** reducing pipeline failures.
- Configured **PII tokenization and column-level encryption** for PHI/Medicaid datasets using KMS and Glue policies.
- Enforced IAM least-privilege roles, VPC endpoints, and network isolation for sensitive workloads.
- **Collaborated with data scientists** to deliver feature-ready datasets that accelerated model development cycles.

Data Engineer – Deloitte USI, India

January 2021 – August 2023

- Designed and migrated large SQL workloads to **AWS Glue, Lambda, and EMR**.
- Built batch & streaming pipelines using **Kafka + AWS lakehouse (S3, Redshift, Iceberg, Glue Catalog)**.
- Implemented **dimensional modeling & star schemas** in Databricks for retail & finance use cases.
- Developed **Python ETL frameworks** supporting schema evolution, partitioning, and large-scale transformations.
- Orchestrated workloads using **Apache Airflow** ensuring SLA-driven, reliable pipelines.

- Developed dashboards using **QuickSight & Tableau**, improving stakeholder access to curated datasets.
- Implemented DR strategies with automated snapshots, cross-AZ backups, and failover procedures.
- **Created ingestion accelerators** to standardize PySpark transformation logic across teams.
- **Integrated Iceberg ACID transactions** improving consistency and rollback safety.
- **Optimized Hive/Spark SQL performance** reducing batch runtime.
- **Implemented event-driven pipelines** using Lambda + S3 triggers for near real-time processing.
- **Built boto3 automation scripts** for RDS, DynamoDB, and S3 data synchronization.
- **Enhanced Glue Catalog governance** with versioning, schema-tracking, and validation rules.
- **Performed deep-dive RCA** to permanently resolve high-priority ETL failures.

Data Engineer – Genius Consultants, India

July 2017 – Dec 2020

- Built ETL pipelines across Oracle, SQL Server, XML, and flat-file sources using **SSIS & Python**.
- Created optimized T-SQL stored procedures, window functions, and CTEs supporting analytics.
- Refactored Spark Streaming pipelines into **Structured Streaming** for modern architecture.
- Automated ETL monitoring & alerting using Shell scripts and SQL Server Agent.
- Developed reconciliation and DQ processes ensuring accurate source-to-target validation.
- **Designed reusable SSIS templates** to speed development and standardize workflows.
- **Implemented CDC-based ETL loads** minimizing full-load operations.
- **Built Python utilities** for file validation, regex parsing, deduplication, and data cleansing.
- **Improved indexing and partitioning strategies**, reducing query runtime.
- **Developed synthetic data generators** for QA and scenario testing.
- **Integrated XML/JSON parsing** into ETL pipelines to support semi-structured ingestion.
- **Automated daily batch cycle monitoring** with error notifications to improve SLA compliance.

EDUCATION

Master of Science – Business Analytics & Artificial Intelligence (May 2025)

University of Texas at Dallas

Bachelor of Science – Computer Science (May 2017)

Adikavi Nannaya University